From Passive Estates to Flexible Generators

A Multi-Vector Digital Twin for Decarbonising Business Parks

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Progress review — the modelling paper (Applied Energy) & the systematic review (RSER)

What I set out to do

- **Question:** can PV + multi-vector storage turn a UK business park from a passive load into a flexible, low-carbon generator?
- **Method:** a physics-based **digital-twin framework** (PyPSA + pvlib + HiGHS) of **Eden Campus**, a real innovation campus in Fife.
- **Approach:** five configurations over **8,760 hours** of optimised dispatch — PV-only (S0) through full hybrid (S4: PV + battery + thermal + hydrogen).
- **Two outputs:** the modelling paper (this review's focus) and a systematic review positioning business parks as a flexibility resource.

The system modelled

Three coupled energy vectors, optimised for least cost:

Vector	Components
Electricity	Solar PV (1 MW), grid import/export, battery (500 kW / 2 MWh)
Hydrogen	Electrolyser → 5 MWh tank → fuel cell (long-duration storage)
Heat	Heat pump (COP 3), PCM thermal store (200 kW / 1.6 MWh), backup boiler

- Weather from committed **PVGIS 2023** (St Andrews, 56.34°N) — reproducible offline.
- Costs on the **capacity-charge tariff basis** (£120/kW/yr, UK commercial DUoS).

Headline result — and a correction I found while preparing this

In the **committed** model, heat was only costed in S2 & S4 — an **unequal comparison**. When I apply the heat demand to **all five** scenarios (fair, like-for-like):

Scenario	Annual cost — committed (£k)	Annual cost — corrected (£k)	Saving vs S0
S0: PV Only	539.0	729.0	—
S1: PV + BESS	468.0	658.1	+70.9
S2: PV + PCM TES	683.3	683.3	+45.7
S3: PV + H_2 LDES	515.6	705.6	+23.4
S4: Full Hybrid	628.4	**628.1**	**+100.9**

Corrected ranking: S4 (full hybrid) is cheapest — not S1. Every storage scenario saves vs the PV-only baseline.

What this changes — and what still holds

Changes (honestly):

- **Cost ranking flips**: S4 cheapest, not S1. The old "S1 optimal" was an artefact of the unequal thermal boundary.
- **Abatement costs**: on total emissions, all storage scenarios still **cost-negative** (MAC −£241 to −£2,155/tCO_2); **S2 abates the most CO_2, S4 is cheapest.**

Still holds (re-verified):

- **S1 still saves £71k/yr vs S0** (both now carry heat).
- **Tariff-dependent heat-pump peak finding** — unchanged (it's electricity-peak only).
- **Demand-robustness** — structural conclusions survive ±30%.

> **I'm bringing this to you as a decision:** agree the fix (incl. boiler sizing) before I re-run the full analysis and update the manuscript.

Two further model issues found (for discussion)

- **Grid export never happens.** Export carries the wrong price sign, so the optimiser never exports — S0 discards ~266 MWh/yr of surplus solar (~26%).
  Fix: correct the export sign to a positive SEG payment.
- **CO_2 factor is dated.** Model uses **0.233 kg/kWh**; the 2023 DESNZ generation factor is **~0.207** — so current CO_2 figures are ~11% high.
  Fix: adopt the chosen DESNZ 2023 factor (decision needed: generation vs consumption).
- **Grid operator corrected.** Fife/Eden Campus is **SP Energy Networks (SPD)**, not SSEN — repo note reversed (SPEN is building a substation at Guardbridge, next to Eden Campus).

The novel finding — the tariff-dependent heat-pump peak

Heat-pump electrification (S2, S4) has **opposite** peak effects depending only on the tariff signal:

- **Energy-only tariff:** cost-minimising dispatch drives peak imports **+35%** (672 → 904 kW) — cheap heat is dispatched without regard to system peak.
- **Capacity/demand charge:** the full hybrid (S4) **reduces** the peak by ~**40%**; S2 stays at baseline (the capacity charge rewards peak management where storage can act on it) — the objective now values peak reduction.

The "objective-function trap"

> A cost-minimising model delivers what the objective *rewards* — not necessarily what the flexibility transition *needs*. The tariff signal must be in the objective.

The honest limitation — demand is synthetic

- Electrical & thermal demand are **representative CIBSE-benchmark profiles**, not metered site data (2,401 MWh/yr, 682 kW peak).
- This is **stated explicitly** in the manuscript as the primary limitation.
- **I have built the pipeline to replace it the moment real data arrives:**
  - `ingest_real_demand.py` — normalises any HH/kWh supplier format
  - `compare_demand_synthetic_vs_real.py` — quantifies how reality differs
  - synthetic baseline preserved; `data_prep.py` swaps in real data, no code change

Why the limitation doesn't change my conclusions

Stress test (re-run on the corrected symmetric-heat model): demand scaled **0.70× – 1.30×** (1,681–3,121 MWh/yr).

Conclusion	Across the whole ±30% band
Cost order **S4 < S1 < S2 < S3 < S0**	**YES — unchanged at every scale**
Every storage option beats PV-only	**YES**
Peak reduction: S1/S4 35–42%, S2/S3 1–12%	**YES**

> Conclusions are **structural**, not artefacts of the demand calibration.
> Real data will refine magnitudes — it will not reverse the findings.

Marginal abatement cost — all storage cost-negative

On the **corrected (symmetric-heat) model, counting total emissions** (grid electricity + boiler gas) vs S0, against the **UK ETS price £49.41/tCO₂**:

Scenario	CO_2 abated vs S0 (t/yr)	MAC (£/tCO₂)
S1: PV + BESS	32.9	−2,155
S2: PV + PCM TES	**189.5 (most)**	−241
S3: PV + H_2	14.8	−1,581
S4: Full Hybrid	156.5	−645 **(cheapest)**

All four abate at negative cost. S2 cuts the most CO_2 (its heat pump displaces ~1,650 MWh of boiler gas); **S4 is the cheapest overall.** No single scenario wins on both — the trade-off is the point.

MAC magnitudes depend on the gas factor (0.183 tCO₂/MWh) and boiler efficiency assumed; ordering and the all-negative result are robust. Source: `verify_total_emissions_mac.py`.

The systematic review (RSER) — first output, well-architected

"From passive estates to flexible generators: a systematic review of UK business parks in electricity markets."

- **Spine:** structural fit → institutional invisibility → objective-function trap.
- **Original contribution:** a 5-type typology (T1–T5) of UK business parks mapped to the FES 2025 flexibility categories — the first such characterisation.
- **Method:** PRISMA 2020 + PRISMA-S, 3 databases + grey corpus, Rayyan, Cohen's κ, with field-specific appraisal (data / scale / market realism).

RSER review — verification status

Verified this session	Status
FES 2025 flexibility figures (127 TWh, 51 GW EV, 10→96 GW storage)	✅ **vs NESO source**
Legal: Class E, Scottish/NI use classes, RED II Art. 22	✅ **primary sources**
Regulatory: TCR, P415, DFS year-round, REMA	✅ **(dates corrected)**
Method refs (PRISMA, Rayyan, Cohen)	✅ **supplied**
Typology exemplars (Edinburgh Park, Eurocentral, MSIP, Catalyst)	✅ **confirmed**

Remaining (the evidence phase): screened-corpus citations & findings, PRISMA flow counts, data-spine park counts — the irreducible systematic-review legwork.

Reproducibility & provenance — built in

- **Every manuscript number traces to a committed artifact:**
  - Tables 3 & 4 → `generate_manuscript_tables.py` (from model output)
  - Table 5 (meteorology) → `table5_meteorological_summary.py` (from raw CSVs)
  - Stress test, MAC, all figures (300 dpi) → committed scripts
- **Single source of truth:** `data_prep.py` — no hand-transcription.
- **All on GitHub**, version-controlled, re-runnable in the `eden_phd` environment.

Where I am and what's next

Done & defensible

- Modelling paper: method sound; **re-run pending agreement on the fixes** (heat boundary, export sign, CO_2 factor)
- Review paper: architected, every verifiable claim checked

Immediate next steps

1. **Ingest real Eden Campus demand** → run comparison harness → re-run model (pipeline ready)
2. Fill RSER review from screened corpus (the evidence phase)
3. Next modelling papers: co-optimised dispatch & peak-aware thermal storage

What I need from you

1. **Agree the model fixes** before I re-run the full analysis:
   - symmetric heat boundary (+ boiler sizing 600→800 kW)
   - export price sign (enable SEG export)
   - CO_2 factor (which DESNZ 2023 figure — generation vs consumption?)
2. **Metered Eden Campus demand** — the single biggest step to validation; the ingest pipeline is ready for it.
3. **A second screener** for the systematic review (dual-screening for Cohen's κ).
4. **Authorship & target-journal confirmation** for both outputs.

Thank you — discussion

Key artifacts (all committed to GitHub):

- `Paper1_Eden_Campus_Manuscript.docx` — modelling paper
- `generate_manuscript_tables.py`, `stress_test_demand_sensitivity.py` — reproducible results
- `paper1_RSER_review/` — review sections + verification records
- `REVIEW_2026-11-07…md` — editorial review (AI-assisted, cross-checked against sources)
- `data_prep.py`, `ingest_real_demand.py`, `compare_demand_synthetic_vs_real.py` — data pipeline

Happy to take questions — including on the synthetic-demand limitation.

Q&A Annex — prep (not for projection)

"Why is S4 the cheapest, not S1?"
On a fair thermal boundary (heat costed in all scenarios), the heat-pump scenarios serve heat at ~£93/MWh (elec ÷ COP 3) vs £110 gas boiler, so S4 ends up cheapest (£628k). The old "S1 optimal" result costed heat only in S2/S4 — an error I found, reproduced, and fixed (`fix_symmetric_heat.py`). S1 still saves £71k/yr vs S0.

"How do I know the fix is right?"
Control run reproduces the committed costs to the penny; the symmetric re-run flips the ranking. Boiler resized 600→800 kW so boiler-only scenarios meet the 756 kW heat peak. All in the repo, re-runnable.

"Your peak finding — 35% or 40%?"
Both, and that's the point: +35% peak *increase* under energy-only tariff; ~40% *reduction* under a capacity charge. Same system, opposite outcomes, tariff-driven. Unaffected by the heat-boundary fix (it's electricity-peak only).

"Why does the model never export?"
A sign error on the export price — found, flagged, quick fix. It means S0 currently discards ~26% of its solar; correcting it will change absolute numbers, not the ranking.

"Isn't this just a simulation, not a digital twin?"
Correct distinction. It's a digital-twin **framework** demonstrated with synthetic